

The artifact selected for this enhancement is a MongoDB based dashboard application originally developed in CS 340: Client/Server Development. The application displays animal shelter data using a table and geolocation map, supported by a Python based module that retrieves and processes data from a MongoDB database. This artifact demonstrates my ability to integrate database systems with user facing applications.

I selected this artifact because it highlights my skills in database management, data retrieval, and data visualization. It demonstrates my ability to work with NoSQL databases, structure queries, and present data in a way that supports meaningful analysis and decision making.

To enhance this artifact, I improved both the functionality and interactivity of the dashboard. I implemented dynamic filtering using radio buttons, allowing users to query data based on rescue categories such as water rescue, wilderness rescue, and disaster tracking. I also added a pie chart visualization to display breed distribution, providing immediate analytical insights. Additionally, I improved the map functionality by enabling users to select entries and dynamically update the displayed location, creating a more intuitive and interactive user experience.

These enhancements demonstrate my ability to design systems that effectively connect backend data with front end visualization tools. I improved data flow, user interaction, and system responsiveness while ensuring that the application remains scalable and maintainable.

From a security standpoint, I considered how data is accessed and presented within the system. By controlling how queries are executed and limiting user interactions to predefined

filters, the application reduces the risk of unintended data exposure and improves data integrity. Additionally, structured data access through a controlled Python module ensures that database interactions remain consistent and secure.

From a stakeholder perspective, this dashboard supports real world decision making by providing clear, interactive insights into animal shelter data. Users can quickly filter, analyze, and visualize information, making the system valuable for operational and analytical purposes. These improvements enhance usability while aligning the system with organizational needs.

Through the enhancement process, I developed a deeper understanding of how database queries, user interface components, and data visualization techniques work together in a full stack application. One challenge I encountered was ensuring that the dashboard updated correctly in response to user interactions while maintaining consistent data flow from the database. I also gained experience improving usability by making the interface more interactive and responsive to user input. The development of this project took place in a virtual lab environment, Codio, which introduced some limitations when testing full application functionality, particularly with external connections and persistent services such as MongoDB. Despite these constraints, core functionality was validated through partial testing within the environment and through prior module work. Additionally, the included project README provides documentation and examples demonstrating the intended functionality of the dashboard and database interactions.

This artifact aligns with program outcomes by demonstrating the ability to design computing solutions using database principles, implement modern tools, and communicate data effectively. Through this process, I strengthened my understanding of how databases, user

interfaces, and visualization techniques work together to deliver meaningful and actionable insights.